

EXTENSION 1

Stage 6

Statistical Analysis (Ext1), S1 The Binomial (Y12)

Statistics and Binomial Distributions (Ext1)

Binomial Probability (Ext1)

Normal Approximations of Sample Proportions (Ext1)

Teacher: Tina Iosifidis

Exam Equivalent Time: 76.5 minutes (based on allocation of 1.5 minutes per mark)

Questions

1. Statistics, EXT1 S1 SM-Bank 4 MC

When a standard 6-sided die is thrown, the probability that it shows a prime number is $\frac{2}{3}$.

If 10 standard dice are thrown, the number, $\mathbf{N}$, of times a prime number is showing has a binomial distribution.

What is the standard deviation of $\mathbf{N}$, correct to 3 decimal places?

- A. 0.222
- B. 0.471
- C. 1.491
- D. 2.222

2. Statistics, EXT1 S1 2021 HSC 6 MC

The random variable $\mathbf{X}$ represents the number of successes in 10 independent Bernoulli trials. The probability of success is $\mathbf{p = 0.9}$ in each trial.

Let $\mathbf{r = P(X \geq 1)}$.

Which of the following describes the value of $\mathbf{r}$?

- A. $\mathbf{r > 0.9}$
- B. $\mathbf{r = 0.9}$
- C. $\mathbf{0.1 < r < 0.9}$
- D. $\mathbf{r \leq 0.1}$

3. Statistics, EXT1 S1 2017 HSC 11g

The probability that a particular type of seedling produces red flowers is $\frac{1}{5}$.

Eight of these seedlings are planted.

- i. Write an expression for the probability that exactly three of the eight seedlings produce red flowers. (1 mark)
- ii. Write an expression for the probability that none of the eight seedlings produces red flowers. (1 mark)
- iii. Write an expression for the probability that at least one of the eight seedlings produces red flowers. (1 mark)

4. Statistics, EXT1 S1 EQ-Bank 20

Netball Australia records show that 10% of all registered players are over the age of 25.

- i. A random survey of 100 netball players was carried out to find out how many were over 25 years of age. Assuming the sample proportion is normally distributed, calculate the expected mean and standard deviation of this group. (2 marks)
- ii. Using the probability table attached, estimate the probability that at least 15 players surveyed will be over 25 years of age. (2 marks)

5. Statistics, EXT1 S1 2018 HSC 12d

A group of 12 people sets off on a trek. The probability that a person finishes the trek within 8 hours is 0.75.

Find an expression for the probability that at least 10 people from the group complete the trek within 8 hours. (2 marks)

6. Statistics, EXT1 S1 SM-Bank 3

In a chocolate factory the material for making each chocolate is sent to a machines.

The time, $\mathbf{X}$ seconds, taken to produce a chocolate by machine is a binomial distribution where it can be shown that $\mathbf{P(X \leq 3) = \frac{9}{32}}$.

A random sample of 10 chocolates is chosen. Find the probability, correct to two decimal places, that exactly 4 of these 10 chocolates took 3 or less seconds to produce. (2 marks)

7. Statistics, EXT1 S1 2023 HSC 11f

A recent census found that 30% of Australians were born overseas.

A sample of 900 randomly selected Australians was surveyed.

Let $\hat{p}$ be the sample proportion of surveyed people who were born overseas.

A normal distribution is to be used to approximate $P(\hat{p} \leq 0.31)$.

- Show that the variance of the random variable $\hat{p}$ is $\frac{7}{30\,000}$. (2 marks)
- Use the standard normal distribution and the normal distribution table to approximate $P(\hat{p} \leq 0.31)$, giving your answer correct to two decimal places. (2 marks)

8. Statistics, EXT1 S1 EQ-Bank 13

On average, batsmen playing cricket in a T20 competition play a scoring shot two out of every three balls.

In a regular season, a total of 1200 overs that each contain six balls, are bowled.

Estimate how many overs would have at least five scoring shots. (3 marks)

9. Statistics, EXT1 S1 SM-Bank 16

For a certain species of bird, the proportion of birds with a crest is known to be $\frac{3}{5}$.

Let $\hat{p}$ be the random variable representing the proportion of birds with a crest in samples of size n for this specific bird.

Find the smallest sample size for which the standard deviation of $\hat{p}$ is less than 0.08. (2 marks)

10. Statistics, EXT1 S1 SM-Bank 5

A manufacturer makes torches that have a probability of 0.03 of being defective.

Let $\hat{p}$ be the random variable that represents the sample proportion of torches for samples of size n drawn from production.

Find the smallest integer value of n such that the standard deviation of $\hat{p}$ is less than $\frac{1}{50}$. (2 marks)

11. Statistics, EXT1 S1 2017 MET1 4

In a large population of fish, the proportion of angel fish is $\frac{1}{4}$.

Let $\hat{p}$ be the random variable that represents the sample proportion of angel fish for samples of size n drawn from the population.

Find the smallest integer value of n such that the standard deviation of $\hat{p}$ is less than or equal to $\frac{1}{100}$. (2 marks)

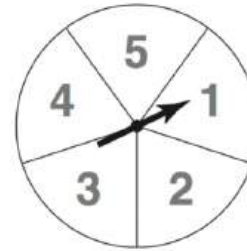
12. Statistics, EXT1 S1 2015 MET2 10

The binomial random variable, X , has $E(X) = 2$ and $\text{Var}(X) = \frac{4}{3}$.

Calculate $P(X = 1)$. (3 marks)

13. Statistics, EXT1 S1 2012 HSC 12c

Kim and Mel play a simple game using a spinner marked with the numbers 1, 2, 3, 4 and 5.



The game consists of each player spinning the spinner once. Each of the five numbers is equally likely to occur.

The player who obtains the higher number wins the game.

If both players obtain the same number, the result is a draw.

- Kim and Mel play one game. What is the probability that Kim wins the game? (1 mark)
- Kim and Mel play six games. What is the probability that Kim wins exactly three games? (2 marks)

14. Statistics, EXT1 S1 2022 HSC 13e

A chocolate factory sells 150-gram chocolate bars. There has been a complaint that the bars actually weigh less than 150 grams, so a team of inspectors was sent to the factory to check. They randomly selected 16 bars, weighed them and noted that 8 bars weighed less than 150 grams.

The factory manager claims 80% of the chocolate bars produced by the factory weigh 150 grams or more.

- The inspectors used the normal approximation to the binomial distribution to calculate the probability, $\mathcal{P}$, of having at least 8 bars weighing less than 150 grams in a random sample of 16, assuming the factory manager's claim is correct.

Using the attached probability table, calculate the value of $\mathcal{P}$. (2 marks)

- The factory manager disagrees with the method used by the inspectors as described in part (i).

Explain why the method used by the inspectors might not be valid. (1 mark)

15. Statistics, EXT1 S1 2021 HSC 14d

At a certain factory, the proportion of faulty items produced by a machine is $p = \frac{3}{500}$, which is considered to be acceptable. To confirm that the machine is working to this standard, a sample of size n is taken and the sample proportion $\hat{p}$ is calculated.

It is assumed that $\hat{p}$ is approximately normally distributed with $\mu = p$ and $\sigma^2 = \frac{p(1-p)}{n}$.

Production by this machine will be shut down if $\hat{p} \geq \frac{4}{500}$.

The sample size is to be chosen so that the chance of shutting down the machine unnecessarily is less than 2.5%.

Find the approximate sample size required, giving your answer to the nearest thousand. (3 marks)

16. Statistics, EXT1 S1 2022 HSC 12e

A game consists of randomly selecting 4 balls from a bag. After each ball is selected it is replaced in the bag. The bag contains 3 red balls and 7 green balls. For each red ball selected, 10 points are earned and for each green ball selected, 5 points are deducted. For instance, if a player picks 3 red balls and 1 green ball, the score will be $3 \times 10 - 1 \times 5 = 25$ points.

What is the expected score in the game? (2 marks)

17. Statistics, EXT1 S1 SM-Bank 8

A laptop's battery is considered faulty if its battery life is less than 3 hours.

The laptop supplier knows that the chance of a faulty battery in any laptop is 6.5%.

A random sample of 70 laptops is selected from the supplier and the battery life of each laptop is tested.

Assuming the sample proportion is normally distributed, what is the probability that the percentage of laptops with faulty batteries lies between 5% and 10%?

Give your answer to the nearest percentage. (3 marks)

18. Statistics, EXT1 S1 2012 MET2 3

Steve and Jess are two students who have agreed to take part in a psychology experiment. Each has to answer several sets of multiple-choice questions. Each set has the same number of questions, n , where n is a number greater than 20. For each question there are four possible options A, B, C or D, of which only one is correct.

a. Steve decides to guess the answer to every question, so that for each question he chooses A, B, C or D at random.

Let the random variable X be the number of questions that Steve answers correctly in a particular set.

i. What is the probability that Steve will answer the first three questions of this set correctly? (1 mark)

ii. Use the fact that the variance of X is $\frac{75}{16}$ to show that the value of n is 25. (1 mark)

b. The probability that Jess will answer any question correctly, independently of her answer to any other question, is p ($p > 0$). Let the random variable Y be the number of questions that Jess answers correctly in any set of 25.

If $P(Y > 23) = 6 \times P(Y = 25)$, show that the value of $p = \frac{5}{6}$. (2 marks)

19. Statistics, EXT1 S1 2022 HSC 14d

An airline company that has empty seats on a flight is not maximising its profit.

An airline company has found that there is a probability of 5% that a passenger books a flight but misses it. The management of the airline company decides to allow for overbooking, which means selling more tickets than the number of seats available on each flight.

To protect their reputation, management makes the decision that no more than 1% of their flights should have more passengers showing up for the flight than available seats.

Given management's decision and using the *attached normal distribution probability table* to find a suitable approximation, find the maximum number of tickets that can be sold for a flight which has 350 seats. (4 marks)

Probability Table for Standard Normal Distribution
Negative z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Probability Table for Standard Normal Distribution
Positive z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Worked Solutions

1. Statistics, EXT1 S1 SM-Bank 4 MC

$$N \sim \text{Bin}(n, p) \sim \text{Bin}\left(10, \frac{2}{3}\right)$$

$$\begin{aligned}\text{Var}(N) &= np(1-p) \\ &= 10 \cdot \frac{2}{3} \left(1 - \frac{2}{3}\right) \\ &= \frac{20}{9}\end{aligned}$$

$$\begin{aligned}\therefore \sigma_N &= \sqrt{\frac{20}{9}} \\ &= 1.4907\dots\end{aligned}$$

$$\Rightarrow C$$

2. Statistics, EXT1 S1 2021 HSC 6 MC

$$p = 0.9, \quad \bar{p} = 0.1$$

$$\begin{aligned}P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - (0.1)^{10} \\ &= 0.999\dots\end{aligned}$$

$$\begin{aligned}\therefore r &> 0.9 \\ \Rightarrow A\end{aligned}$$

♦ Mean mark 50%.

Worked Solutions

3. Statistics, EXT1 S1 2017 HSC 11g

$$\begin{aligned}\text{i. } P(\text{Red}) &= \frac{1}{5}, \quad P(\text{Not Red}) = \frac{4}{5} \\ P(\text{exactly 3 are red}) \\ &= {}^8C_3 \cdot \left(\frac{1}{5}\right)^3 \cdot \left(\frac{4}{5}\right)^5\end{aligned}$$

$$\begin{aligned}\text{ii. } P(\text{none are red}) \\ &= {}^8C_0 \cdot \left(\frac{1}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^8 \\ &= \left(\frac{4}{5}\right)^8\end{aligned}$$

$$\begin{aligned}\text{iii. } P(\text{at least 1 is red}) \\ &= 1 - P(\text{none are red}) \\ &= 1 - \left(\frac{4}{5}\right)^8\end{aligned}$$

4. Statistics, EXT1 S1 EQ-Bank 20

$$\begin{aligned}\text{i. } E(\hat{p}) &= p = 0.1 \\ \text{Var}(\hat{p}) &= \frac{p(1-p)}{n} \\ &= \frac{0.1(0.9)}{100} \\ &= 0.0009 \\ \sigma(\hat{p}) &= \sqrt{0.0009} \\ &= 0.03\end{aligned}$$

$$\begin{aligned}\text{ii. } P(\text{at least 15 players are over 25}) &= P(\hat{p} \geq 0.15) \\ \hat{p} &\sim N(\mu, \sigma) \sim N(0.1, 0.03) \\ P(\hat{p} \geq 0.15) &= P\left(z \geq \frac{0.15 - 0.10}{0.03}\right) \\ &= P(z \geq 1.67) \\ &= 1 - 0.9525 \\ &= 0.0475\end{aligned}$$

5. Statistics, EXT1 S1 2018 HSC 12d

$$P(\text{finishes within 8 hours}) = 0.75$$

$$P(\text{not finish within 8 hours}) = 0.25$$

Let X = number who finish below 8 hours

$$\therefore P(\text{at least 10 finish within 8 hours})$$

$$= P(X=10) + P(X=11) + P(X=12)$$

$$= \binom{12}{10} (0.75)^{10} (0.25)^2 + \binom{12}{11} (0.75)^{11} (0.25)^1 + \binom{12}{12} (0.75)^{12}$$

6. Statistics, EXT1 S1 SM-Bank 3

Let C = number of chocolates that take less than 3 seconds

$$C \sim \text{Bin}(n, p) \sim \text{Bin}\left(10, \frac{9}{32}\right)$$

COMMENT: Take care as $P(X \leq 3) = \frac{9}{32}$ provides the equivalent of p here.

$$P(C = 4) = \binom{10}{4} \left(\frac{9}{32}\right)^4 \left(\frac{23}{32}\right)^6$$

$$= 0.181\dots$$

$$= 0.18 \quad (2 \text{ d.p.})$$

7. Statistics, EXT1 S1 2023 HSC 11f

i. Let X = number of people born overseas

$$X \sim \text{Bin}(900, 0.3)$$

$$\hat{p} = \frac{X}{900}$$

$$\text{Var}(X) = np(1-p) = 900 \times 0.3 \times 0.7$$

$$\text{Var}(\hat{p}) = \frac{\text{Var}(X)}{n^2}$$

$$= \frac{900 \times 0.3 \times 0.7}{900^2}$$

$$= \frac{7}{30\,000}$$

$$\text{ii. } \mu = E(\hat{p}) = 0.3, \quad \sigma(\hat{p}) = \sqrt{\frac{7}{30\,000}}$$

$$z\text{-score}(0.31) = \frac{0.31 - 0.3}{\sqrt{\frac{7}{30\,000}}} = 0.6546\dots$$

$$P(\hat{p} \leq 0.31) = P(z \leq 0.6546\dots)$$

$$= 0.74 \quad (\text{to 2 d.p.})$$

8. Statistics, EXT1 S1 EQ-Bank 13

Let X = number of scoring shots in a six ball over

$$X \sim \text{Bin}\left(6, \frac{2}{3}\right)$$

$$P(X \geq 5) = P(X = 5) + P(X = 6)$$

$$= {}^6C_5 \cdot \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) + {}^6C_6 \cdot \left(\frac{2}{3}\right)^6$$

$$= \frac{64}{243} + \frac{64}{729}$$

$$= \frac{256}{729}$$

Let Y = number of overs with at least 5 scoring shots

$$Y \sim \text{Bin}\left(1200, \frac{256}{729}\right)$$

$$E(Y) = np$$

$$= 1200 \times \frac{256}{729}$$

$$= 421.39\dots$$

$$= 421 \quad (\text{nearest over})$$

9. Statistics, EXT1 S1 SM-Bank 16

$$E(\hat{p}) = p = 0.6$$

$$\text{Var}(\hat{p}) = \frac{0.6(1-0.6)}{n} = \frac{0.24}{n}$$

$$\sigma(\hat{p}) = \sqrt{\frac{0.24}{n}}$$

$$\text{If } \sigma(\hat{p}) < 0.08,$$

$$\sqrt{\frac{0.24}{n}} < 0.08$$

$$0.0064n > 0.024$$

$$n > \frac{0.24}{0.0064}$$

$$n > 37.5$$

$$\therefore \text{Smallest } n = 38$$

10. Statistics, EXT1 S1 SM-Bank 5

Let X = number of defective torches

$$X \sim \text{Bin}(n, p) \sim \text{Bin}(n, 0.03)$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$\begin{aligned}\sigma(\hat{p}) &= \sqrt{\frac{0.03(0.97)}{n}} \\ &= \sqrt{\frac{291}{10\,000n}}\end{aligned}$$

$$\sqrt{\frac{291}{10\,000n}} < \frac{1}{50}$$

$$\frac{291}{10\,000n} < \frac{1}{2500}$$

$$10\,000n > 2500 \times 291$$

$$n > \frac{2500 \times 291}{10\,000}$$

$$> 72.75$$

$$\therefore n_{\min} = 73$$

11. Statistics, EXT1 S1 2017 MET1 4

Let X = number of angel fish

$$X \sim \text{Bin}\left(n, \frac{1}{4}\right)$$

$$\begin{aligned}\sigma(\hat{p}) &= \sqrt{\frac{p(1-p)}{n}} \\ &= \sqrt{\frac{\frac{1}{4} \times \frac{3}{4}}{n}} \\ &= \sqrt{\frac{3}{16n}}\end{aligned}$$

$$\text{Solve: } \sqrt{\frac{3}{16n}} \leq \frac{1}{100}$$

$$\frac{3}{16n} \leq \frac{1}{10\,000}$$

$$\frac{30\,000}{16} \leq n$$

$$\therefore n \geq 1875$$

$$\therefore n_{\min} = 1875$$

12. Statistics, EXT1 S1 2015 MET2 10

$$np = 2 \dots (1)$$

$$np(1-p) = \frac{4}{3} \dots (2)$$

Solve simultaneous equations:

Substitute $np = 2$ into (2)

$$2(1-p) = \frac{4}{3}$$

$$2 - 2p = \frac{4}{3}$$

$$p = \frac{1}{3}$$

$$n = 6, \quad p = \frac{1}{3}$$

$$\therefore X \sim \text{Bin}\left(6, \frac{1}{3}\right)$$

$$\therefore P(X=1) = \binom{6}{1} \times \left(\frac{1}{3}\right)^1 \times \left(\frac{2}{3}\right)^5$$

$$= 6 \times \frac{1}{3} \times \left(\frac{2}{3}\right)^5$$

$$= \frac{64}{243}$$

13. Statistics, EXT1 S1 2012 HSC 12c

i. Method 1

$$P(\text{Kim wins game})$$
$$= P(\text{K spins 5}) \times P(\text{M} < 5) + P(\text{K spins 4}) \times P(\text{M} < 4) + \dots$$
$$= \left(\frac{1}{5} \times \frac{4}{5}\right) + \left(\frac{1}{5} \times \frac{3}{5}\right) + \left(\frac{1}{5} \times \frac{2}{5}\right) + \left(\frac{1}{5} \times \frac{1}{5}\right)$$
$$= \frac{10}{25}$$
$$= \frac{2}{5}$$

Method 2

$$P(\text{Draw}) = \frac{1}{5}$$
$$\therefore P(\text{Not a draw}) = 1 - \frac{1}{5} = \frac{4}{5}$$

Since Kim and Mel have equal chance

$$P(\text{K wins}) = \frac{1}{2} \times \frac{4}{5}$$
$$= \frac{2}{5}$$

ii. $P(\text{Kim wins}) = \frac{2}{5}$

$$P(\text{Kim doesn't win}) = \frac{3}{5}$$

After 6 games,

$$P(\text{Kim wins exactly 3})$$
$$= {}^6C_3 \left(\frac{2}{5}\right)^3 \left(\frac{3}{5}\right)^3$$
$$= \frac{6!}{3!3!} \times \frac{8}{125} \times \frac{27}{125}$$
$$= \frac{864}{3125}$$

♦ Mean mark 37%

14. Statistics, EXT1 S1 2022 HSC 13e

i. Let X = number of chocolate bars < 150 grams

$$X \sim \text{Bin}(16, 0.2)$$
$$\mu = np = 16 \times 0.2 = 3.2$$
$$\sigma^2 = np(1 - p) = 16 \times 0.2 \times 0.8 = 2.56$$
$$z\text{-score } (X = 8) = \frac{8 - 3.2}{\sqrt{2.56}} = 3$$

$$\mathcal{P}(X \geq 8) = P(z \geq 3.0)$$
$$= 1 - P(z \leq 3.0)$$
$$= 1 - 0.9987$$
$$= 0.0013$$

ii. $np = 3.2, \quad n(1 - p) = 16 \times 0.8 = 12.8$
Since $np < 5$, the sample size is too small.

♦ Mean mark part (i) 37%.
Mean mark part (ii) 54%.

15. Statistics, EXT1 S1 2021 HSC 14d

$$E(\hat{p}) = p = \frac{3}{500} = 0.006$$
$$P\left(\hat{p} \geq \frac{4}{500}\right) < 2.5\%$$
$$\Rightarrow P\left(\hat{p} \geq \frac{4}{500}\right) = P(z\text{-score} \leq 2)$$
$$2 \times \sigma = \frac{4}{500} - E(\hat{p})$$
$$= \frac{4}{500} - \frac{3}{500}$$
$$= \frac{1}{500}$$
$$\sigma = \frac{1}{1000}$$

Using $\sigma^2 = \frac{p(1-p)}{n}$
$$0.001^2 = \frac{0.006(0.994)}{n}$$
$$n = \frac{0.006(0.994)}{0.001^2}$$
$$= 5964$$
$$\approx 6000$$

♦♦ Mean mark 27%.

16. Statistics, EXT1 S1 2022 HSC 12e

Let X = number of red balls selected

$$X \sim \text{Bin}(4, 0.3)$$

$$\text{Score } (X = 4) = (0.3)^4 \times 40 = 0.324$$

$$\text{Score } (X = 3) = \binom{4}{3} (0.3)^3 (0.7) \times 25 = 1.89$$

$$\text{Score } (X = 2) = \binom{4}{2} (0.3)^2 (0.7)^2 \times 10 = 2.646$$

$$\text{Score } (X = 1) = \binom{4}{1} (0.3) (0.7)^3 \times -5 = -2.058$$

$$\text{Score } (X = 0) = (0.7)^4 \times -20 = -4.802$$

$$\begin{aligned} \therefore E(X) &= 0.324 + 1.89 + 2.646 - 2.058 - 4.802 \\ &= -2 \end{aligned}$$

♦♦ Mean mark 35%.

17. Statistics, EXT1 S1 SM-Bank 8

$$E(\hat{p}) = p = 0.065$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$\begin{aligned} \sigma^2(\hat{p}) &= \frac{0.065(1-0.065)}{70} \\ &= 0.0008682\dots \end{aligned}$$

$$\sigma(\hat{p}) = 0.029465\dots$$

$$\hat{p} \sim N(\mu, \sigma) \sim N(0.065, 0.029465)$$

$$\begin{aligned} z\text{-score } (5\%) &\approx \frac{0.05-0.065}{0.029465} \\ &\approx -0.509 \end{aligned}$$

$$\begin{aligned} z\text{-score } (10\%) &\approx \frac{0.10-0.065}{0.029465} \\ &\approx 1.187 \end{aligned}$$

Using probability tables (attached):

$$\begin{aligned} \therefore P(5\% \leq \hat{p} \leq 10\%) &\approx P(-0.51 \leq z \leq 1.19) \\ &\approx 0.8830 - 0.3050 \\ &\approx 0.5780 \\ &\approx 58\% \end{aligned}$$

18. Statistics, EXT1 S1 2012 MET2 3

$$\begin{aligned} \text{a.i. } P(3 \text{ correct in a row}) &= \left(\frac{1}{4}\right)^3 \\ &= \frac{1}{64} \end{aligned}$$

$$\text{a.ii. } \text{Var}(X) = np(1-p)$$

$$\frac{75}{16} = n \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)$$

$$75 = 3n$$

$$\therefore n = 25$$

$$\text{b. } Y \sim \text{Bin}(25, p)$$

$$P(Y > 23) = 6 \times P(Y = 25)$$

$$P(Y = 24) + P(Y = 25) = 6 \times P(Y = 25)$$

$$P(Y = 24) = 5 \times P(Y = 25)$$

$$\binom{25}{24} p^{24} (1-p)^1 = 5p^{25}$$

$$25p^{24} (1-p) = 5p^{25}$$

$$25p^{24} - 25p^{25} - 5p^{25} = 0$$

$$25p^{24} - 30p^{25} = 0$$

$$5p^{24} (5-6p) = 0$$

$$\therefore p = \frac{5}{6}, \quad (p > 0) \quad \dots \text{ as required}$$

♦♦♦ Mean mark part (c) 19%.

19. Statistics, EXT1 S1 2022 HSC 14d

Let X = number of passengers taking a flight

$$X \sim \text{Bin}(n, 0.95)$$

$$E(X) = 0.05, \quad \text{Var}(X) = n(0.95)(1 - 0.95) = 0.0475n$$

X can be approximated by $Y \sim N(0.95n, 0.0475n)$

Find n such that $P(Y > 350) = 0.01$:

Using the probability table

$\Rightarrow$ z -score of 2.33 corresponds to (closest) upper tail probability < 0.01

$$2.33 = \frac{350 - 0.95n}{\sqrt{0.0475n}}$$

$$2.33\sqrt{0.0475}\sqrt{n} = 350 - 0.95n$$

$$0.95n + 2.33\sqrt{0.0475}\sqrt{n} - 350 = 0$$

$$\begin{aligned} \sqrt{n} &= \frac{-2.33\sqrt{0.0475} \pm \sqrt{\left(2.33\sqrt{0.0475}\right)^2 - 4(0.95)(-350)}}{2(0.95)} \\ &= 18.9288\dots \quad (n > 0) \\ &= (18.9288\dots)^2 \\ &\approx 358.30 \end{aligned}$$

$\therefore$ Maximum tickets that can be sold = 358

◆◆◆ Mean mark 22%.